

Unit 3, Lesson 6: Using Angle Relationships, Postulates, and Theorems



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



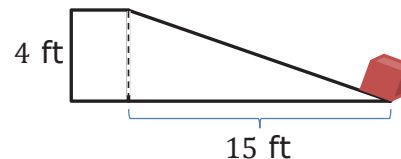
Learning Target

- ☐ I can solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

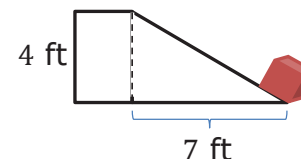


3.6.1 Warm-Up: Would You Rather?

- Two ramps set up to move boxes up to a 4 foot (ft) high platform are shown.



Ramp A



Ramp B

- Would you rather push a 50 pound (lb) box up Ramp A or Ramp B?
- Justify your choice.



3.6.2 Activity: Solving Using Angle Relationships, Postulates, and Theorems

With your partner, sort the cards to match each image with the type of angle pair represented and the value of x . Record your answers in the table, showing your work in the last column.

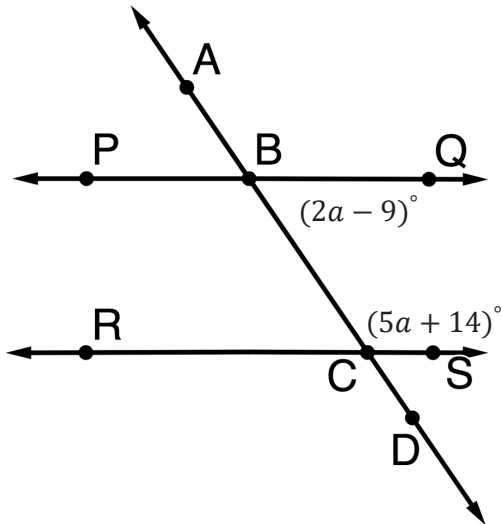
Type of Angle Pair	Letter of Image	Work and Answer

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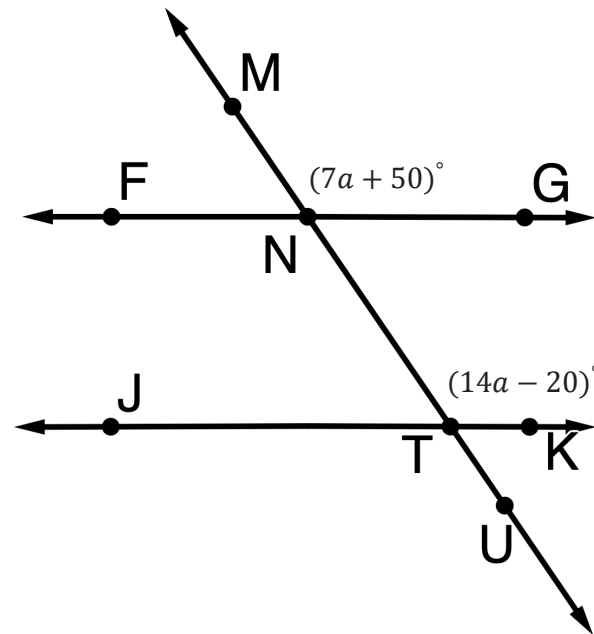
3.6.3 Your Turn!

1. $\overleftrightarrow{PQ} \parallel \overleftrightarrow{RS}$, where $\overleftrightarrow{AD}$ is a transversal.



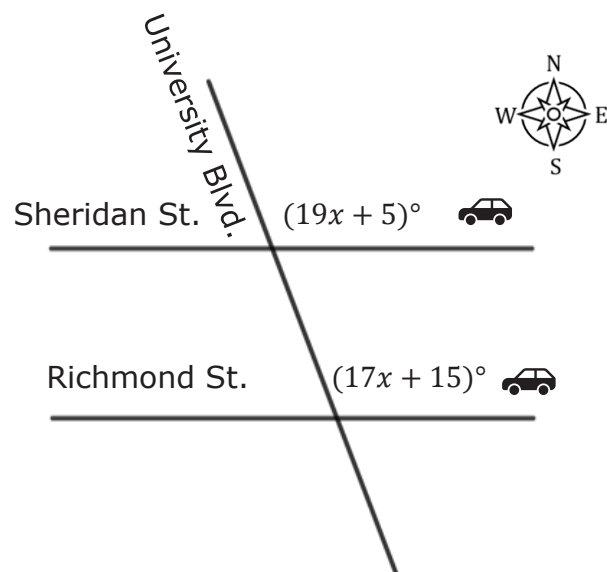
- Determine the value of a .
- What is $m\angle ABQ$?
- What is $m\angle BCR$?

2. $\overleftrightarrow{FG} \parallel \overleftrightarrow{JK}$ where $\overleftrightarrow{MU}$ is a transversal.



- Determine the value of a .
- What is $m\angle FNT$?
- What is $m\angle KTU$?

3. Sheridan Street and Richmond Street are parallel, and both are intersected by University Boulevard.

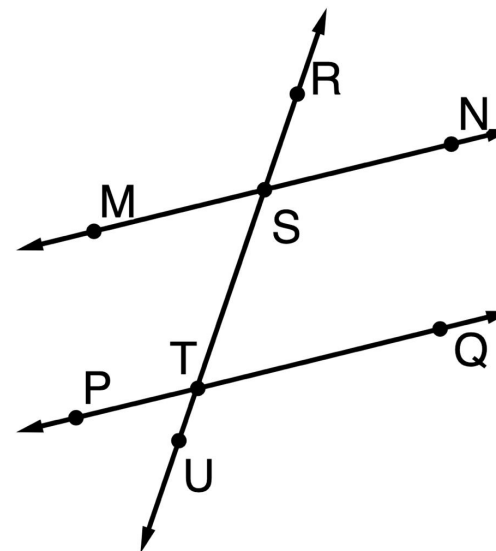


Two cars are heading west, one on Sheridan St. and one on Richmond St. Both cars turn north onto University Blvd. What is the measure of the angle of their turn onto University Blvd.?



3.6.4 Wrap-Up: Ticket Out the Door

1. Given $\overleftrightarrow{MN} \parallel \overleftrightarrow{PQ}$, $m\angle PTS = (14x - 10)^\circ$, and $m\angle NST = (9x + 35)^\circ$, find each angle measure.



a. $m\angle MSR$

b. $m\angle STQ$